

Solution Requirements

Date	4 july,2026
Team ID	SWTID-2026-5594
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system allows users to securely access the application through the web interface. (If login is not implemented, mention that the system is accessible for demo purposes.)</i>	High
2	Authorization levels	<i>Users can access the prediction module. Administrators can manage the trained model and application files.</i>	Medium
3	External interfaces	<i>The system uses a trained Machine Learning model (credit_card_model.pkl) and processes input through the Flask web application.</i>	High

4	Transactions processing	<i>The application accepts applicant information, processes it using the ML model, and generates an approval or rejection prediction instantly.</i>	High
5	Reporting	<i>Displays prediction results and dashboard statistics. The result page clearly shows whether the application is approved or rejected.</i>	Medium
6	Business rules, etc.	<i>The prediction is based on applicant information such as income, employment, education, family details, and other financial attributes processed by the trained ML model.</i>	High
7	Compliance to laws or regulations	<i>Applicant information should be handled securely and used only for prediction purposes while protecting user privacy.</i>	Medium
8	Other	<i>The application provides a simple, responsive, and user-friendly interface for easy prediction and demonstration.</i>	LOW

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The application should process user input and display the prediction quickly.</i>	<i>Prediction displayed within 2–3 seconds</i>

2	Scalability	<i>The system should support multiple prediction requests without affecting performance</i>	<i>Supports multiple users with stable performance</i>
3	Security & Data Privacy	<i>Applicant information should be processed securely and not shared with unauthorized users.</i>	<i>Secure handling of user input and prediction data</i>
4	Reliability & Availability	<i>The web application should remain available during normal usage and produce consistent prediction results.</i>	99% application availability during operation
5	Usability & Accessibility	<i>The application should provide a simple, responsive, and easy-to-use interface for all users.</i>	<i>User-friendly interface compatible with desktop and mobile browsers</i>
6	<i>Other</i>	<i>The application should be easy to maintain, update, and deploy in different environments such as local systems or cloud platforms.</i>	Maintainable and portable architecture